Abdelhak Kadouma

Software Engineer

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Professional Summary

Software Engineer with over three years of industry experience building and deploying production web applications, backend services, APIs, integrations, and AI-enabled systems. Experienced across Python, Java, JavaScript/TypeScript, Node.js, React, Vue, FastAPI, Django, SQL databases, Docker, and Kubernetes. Proven experience independently delivering production platforms from end to end, leading development of distributed cloud-edge software, and building LLM and machine-learning systems. Research experience adds depth in forecasting, drift detection, model evaluation, and reproducible AI experimentation.

Technical Skills

Software Engineering: Python, Java, JavaScript, TypeScript, Node.js, FastAPI, Django, Spring Boot, React, Vue, HTML, CSS, REST APIs, GraphQL, microservices
AI Engineering: LLM integration, structured outputs, asynchronous AI workflows, scikit-learn, PyTorch, time-series forecasting, drift detection, model evaluation
Data and Databases: PostgreSQL, MySQL, MariaDB, MongoDB, Redis, SQLite, Pandas, NumPy, data ingestion, transformation, validation
Cloud and DevOps: Docker, Kubernetes, DigitalOcean, IONOS Cloud, Git, CI/CD, automated testing
Reliability and Observability: Structured logging, health checks, runtime monitoring, Prometheus, Grafana, debugging, service integration

Professional Experience

Fingletek Oy Sep. 2023 – Present
Software Engineer and Research Collaborator *Espoo, Finland*

- Design, build, deploy, and maintain production applications spanning responsive frontend interfaces, backend services, REST and GraphQL APIs, databases, integrations, and cloud infrastructure using JavaScript, TypeScript, React, Vue, Python, Java, and Node.js.
- Led frontend and backend development of the Ontology Semantic Reasoner (OSR), a major component of the European AC3 cloud-edge framework, integrating semantic application information with Piveau Hub through GraphQL and translating end-user requirements into deployable Kubernetes manifests enriched with data-source information.
- Build AI-enabled services with FastAPI, asynchronous processing, structured-output validation, persistent job tracking, and downstream application integrations.
- Improve production reliability across Docker, Kubernetes, DigitalOcean, IONOS Cloud, and company-managed infrastructure through CI/CD, automated testing, Nginx, health checks, structured logging, runtime monitoring, and debugging across APIs, databases, networking, and service integrations.

Selected Projects

9jashares – Financial Trading Platform 2024 – 2026

- Independently designed, built, deployed, and maintained a production financial web platform from scratch, delivering responsive interfaces, backend workflows, APIs, persistence, integrations, and production deployment end to end.

OSR – Ontology Semantic Reasoner 2023 – 2026

- Led full-stack development of a core AC3 CECCM component using Node.js, GraphQL, semantic application information, SQLite-based authentication, and MySQL/MariaDB profile storage to transform end-user requirements into Kubernetes deployment manifests with associated data-source information.

DynAPM – Drift-Aware Profile Maintenance 2025 – 2026

- Developed a forecasting and online drift-detection pipeline evaluated on traces from 1,576 production virtual machines, improving CPU and memory profile-class accuracy by 13.0 and 14.5 percentage points and reducing SLA-risk events by 28%.

Education

Higher School of Computer Science 2017 – 2023
State Engineering Degree and Master's Degree in Computer Science *Algiers, Algeria*

Languages

Arabic: Native | **English:** Professional working proficiency | **French:** Professional working proficiency